s9c_inverse_vol_filtered

equity strategy | family: equity-volatility | status: research | derived from `s6_neutral_levered`

What this strategy does

Inverse-vol sizing combined with the walk-forward cell filter

Why it should work. Skip the cells that history says do not pay -- but the data shows a gradient, not a bad cell.

Design notes (from the strategy's source file).

Strategy 9a / 9c -- walk-forward volatility-cell filter (alone and combined with inverse-vol).

Drops a (side, idio-vol tercile) cell from new entries when its own history says it has not earned its keep, judged ONLY on quarters whose positions had fully exited. Result: filtering does not help on its own (Sharpe 1.23 vs 1.30 baseline-equivalent run; the effect is a smooth gradient, not one broken cell) and adds nothing once inverse-vol sizing is on (9c 1.41 < 9b 1.47). Kept as a documented negative result: size by vol, do not filter by it.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

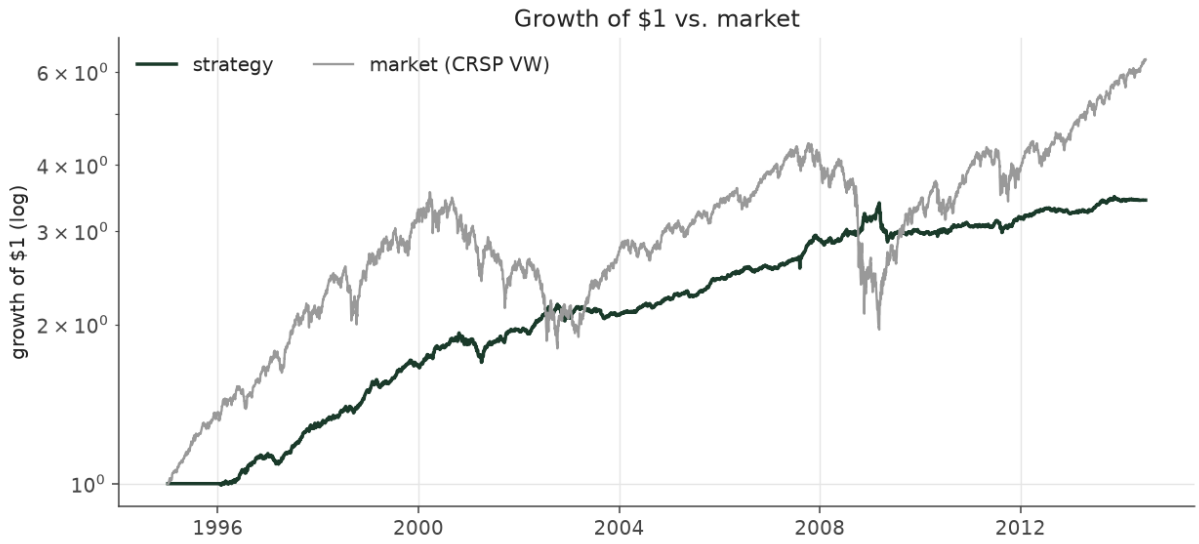
metric	s9c inverse vol filtered	market (CRSP VW)
Annualized return	6.54%	9.94%
Annualized volatility	4.55%	19.35%
Sharpe (NAV-pct)	1.41	0.59
Sortino	2.07	
Max drawdown	-15.7%	-55.5%
Calmar	0.42	
Final NAV ($10M start)	$34,340,070	
Worst day	-3.42%	
% positive days	52.3%	
Sharpe 1996-2001	2.26	
Sharpe 2002-2007	1.57	
Sharpe 2008-2013	0.60	

Correlation of daily returns with the market: **0.02**.

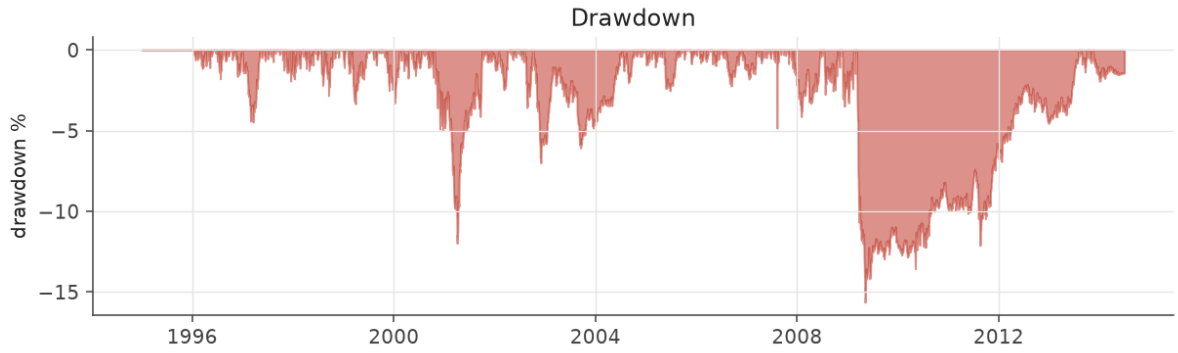
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 1.41

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.0654

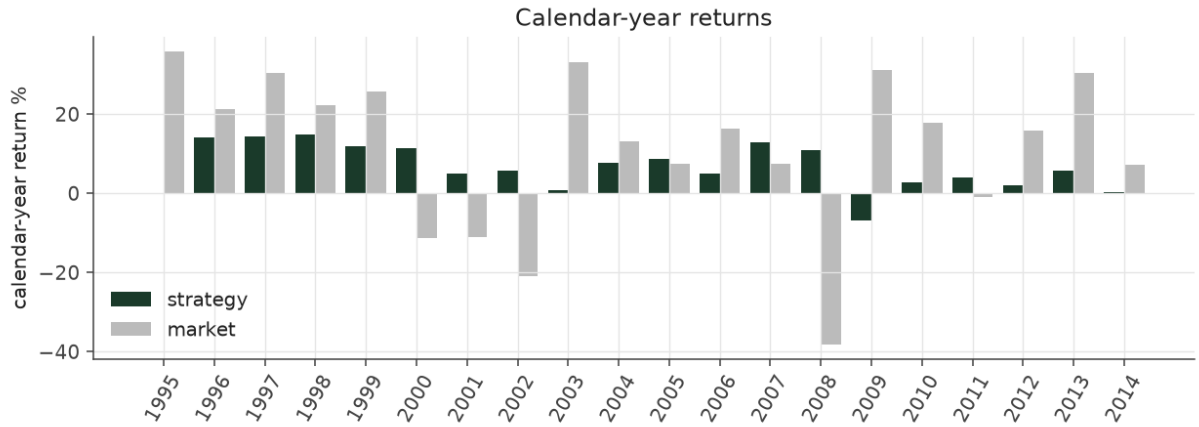
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.418



NAV (log scale)



Drawdown from running peak

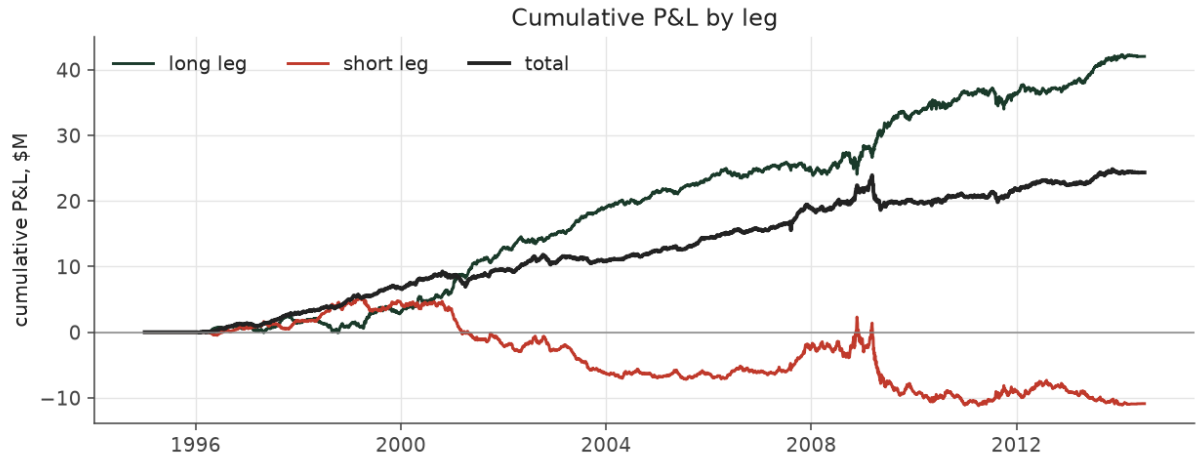


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	42.03	-10.89
P&L after costs ($M)	38.76	-14.42
of which common (average-event) component ($M)	26.55	-27.19
of which excess / PEAD component ($M)	15.48	16.30
excess minus costs ($M)	12.21	12.77
gross return per $ traded (bp)	253.44	-64.00
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.33	-0.25

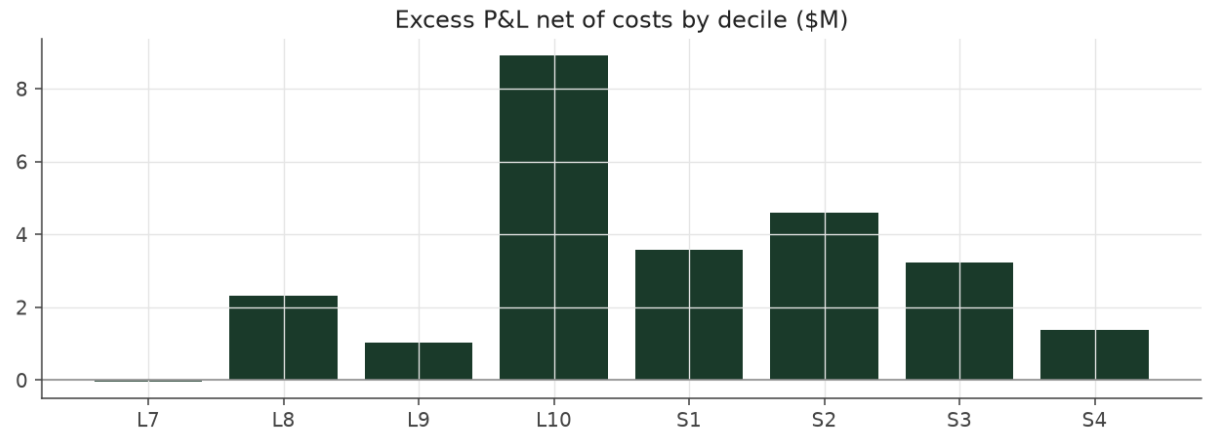
Daily P&L correlation between the two legs: **-0.81**.

• **The short leg loses money in absolute terms** (-10.9M before costs) while the long leg earns +42.0M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 26.5M / 27.2M. Stripping that common component, the surprise signal itself earned +15.5M on the long side and +16.3M on the short side.

• The two legs' daily P&L are -0.81 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.33 long / -0.25 short vs 1.41 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,009	146.90	2.71	0.24	0.29	-0.05	-0.2%	-3.48
long	8	16,508	390.71	8.78	3.09	0.78	2.31	9.3%	59.15
long	9	17,676	513.54	8.55	2.04	1.00	1.04	4.2%	20.23
long	10	17,111	607.20	18.71	10.12	1.20	8.91	35.7%	146.75
short	1	17,093	608.49	-5.44	4.89	1.31	3.58	14.3%	58.83
short	2	17,204	526.88	-4.28	5.68	1.08	4.60	18.4%	87.37
short	3	15,897	402.46	-3.39	4.04	0.82	3.22	12.9%	80.05
short	4	9,154	163.49	-1.32	1.68	0.31	1.37	5.5%	83.62



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,166	294.58	5.45	7.32	1.14	6.18	24.7%	209.69
2	22,785	622.38	5.57	7.68	1.77	5.91	23.7%	94.96
3	25,396	769.16	6.24	7.74	1.64	6.10	24.4%	79.32
4	27,473	832.21	4.17	5.35	1.32	4.02	16.1%	48.36
5	28,832	841.35	2.90	3.70	0.93	2.77	11.1%	32.91

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,618	225.32	5.01	6.56	1.13	5.43	21.7%	241.08
2	23,010	687.46	5.62	8.59	2.06	6.53	26.1%	94.95
3	25,773	838.97	6.51	8.59	1.68	6.91	27.7%	82.40
4	27,539	815.59	4.56	4.99	1.14	3.85	15.4%	47.23

5	28,707	792.06	2.61	3.01	0.79	2.22	8.9%	28.05

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,693	1,290.11	0.52	7.71	2.30	5.41	21.7%	41.96
41-60d	25,536	768.41	4.74	4.86	1.58	3.28	13.1%	42.65
61-90d	35,163	967.77	7.01	8.50	2.07	6.43	25.8%	66.48
>90d	12,260	333.38	12.07	10.71	0.85	9.86	39.5%	295.65

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,652	683.38	9.27	10.83	1.35	9.47	37.9%	138.65
2002-07	39,722	1,115.51	9.49	11.69	2.21	9.47	37.9%	84.93
2008-13	45,278	1,560.78	5.58	9.27	3.24	6.03	24.1%	38.65

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,907	331.60	1.91	2.60	0.71	1.89	7.6%	57.02
Chems	2,587	224.04	1.36	1.74	0.40	1.35	5.4%	60.14
Durbl	2,836	241.66	2.31	2.81	0.44	2.37	9.5%	98.02
Enrgy	6,186	271.01	3.68	4.29	0.51	3.78	15.1%	139.41
Hlth	11,622	321.02	2.54	3.22	0.68	2.54	10.1%	78.96
Manuf	12,007	300.44	1.79	2.59	0.63	1.96	7.9%	65.30
Money	17,569	293.98	2.56	3.33	0.67	2.66	10.6%	90.40
NoDur	5,074	270.67	1.34	1.84	0.52	1.32	5.3%	48.90
Other	18,309	319.76	2.56	3.37	0.72	2.65	10.6%	82.71
Shops	11,943	311.85	3.09	3.75	0.65	3.10	12.4%	99.33
Telcm	3,942	263.04	0.64	1.13	0.48	0.65	2.6%	24.59
Utils	4,670	210.60	0.56	1.12	0.39	0.73	2.9%	34.63

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,046	814.48	3.66	6.05	1.51	4.54	18.2%	55.81
low vol	40,091	1,402.43	11.45	14.69	3.03	11.65	46.7%	83.11
mid vol	37,903	1,094.45	9.05	10.73	2.16	8.56	34.3%	78.26

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	6.54%	0.00%
0.5%	1.7	6.26%	0.28%
1.0%	3.4	5.96%	0.57%
2.0%	6.9	5.33%	1.21%
5.0%	17.2	2.81%	3.73%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.